BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 24

Note: Note: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Mark.

Q1. 1. Choose the correct Answer for each from the given question:

Q1. Blood is one of the following:

(A) Mixture
(B) Compound
(C) Element
(D) Metal

Q2. Empirical formula of CH3COOH is: : CH3COOH

(A) C2H4O2
(B) CH2O
(C) CH2O
(D) C2H4O2

Q3. It is considered as the fourth state of matter:

(A) Solid
(B) Liquid
(C) Gas
(D) Plasma

Q4. The mass of this particle is almost equal to that of proton:

(A) Electron
(B) Neutron
(C) α – Particles
(D) β – particles

Q5. Line spectrum of Hydrogen atom was studied by:

(A) Neil Bohr
(B) Albert Einstein
(C) De Broglie
(D) Max Planck

Q6. This is the isotope of Hydrogen which contains greater number of : neutron:

(A) Protium
(B) Deuterium
(C) Tritium
(D) Hydrogen

Q7. The radioactive element in 1-A group is: : A-1

(A) Rubidium
(B) Francium
(C) Cesium
(D) Potassium

Q8. Dobereiner arranged elements in:

(A) Octaves
(B) Triads
(C) Groups and Periods
(D) Sub-Groups

Q9. All transition elements are:

(A) Gases
(B) Metals
(C) Non-Metals
(D) Metalloids

Q10. According to the modern periodic law presented by Moseley, the properties of elements are the periodic function of:

(A) Atomic Size
(B) Atomic Mass :
(C) Atomic Diameter
(D) Atomic Number

Q11. Synthetic resin binds two surfaces together by this bonding:

(A) Hydrogen Bonding
(B) Ionic Bonding
(C) Covalent Bonding
(D) Metallic Bonding

Q12. Nobel gases are stable because their valance shell contains these numbers of electron: (except Helium)

(A) 4
(B) 5 (:)
(C) 8
(D) 10 6

Q13. At room temperature, most of the ionic compounds are:

(A) Amorphous Solid
(B) Crystalline Solid
(C) Liquid
(D) Gas

Q14. 1 atm = _____ Pascal 1 atm = _____ Pascal

(A) 10325
(B) 106075
(C) 106075
(D) 10325

Q15. Which of the following gas diffuses faster?

(A) Oxygen
(B) Chlorine
(C) Fluorine
(D) Helium

Q16. Benzene is an example of this type of solvent:

(A) Polar
(B) non polar
(C) ionic
(D) universal

Q17. 69 g of Sodium (Na) is equal to: : (Na) 69 g

(A) 1 moles
(B) 2 moles
(C) 2
(D) 1

Q18. The size of particles in suspension is: : (Suspension)

(A) 103nm
(B) 102nm
(C) 102nm
(D) 103nm

Q19. This one of the following is non-electrolyte:

(A) Aqueous HCl
(B) Urea
(C) HCl
(D) Aqueous NaCl

Q20. This one is an alloy:

(A) Graphite
(B) Steel
(C) Mercury
(D) Water

Q21. 1 g equivalent weight of Al is equal to: : Al

(A) 1g
(B) 9g
(C) 9g
(D) 1g

Q22. This gas turns lime water milky :

(A) CO2
(B) O2
(C) O2
(D) CO2

Q23. This halogen exists in liquid state:

(A) Fluorine
(B) Chlorine
(C) Iodine
(D) Bromine

Q24. This element is metalloid: : (Metalloid)

(A) Br
(B) Si
(C) Si
(D) Br

SECTION “B” : (Short Answer Question)

Maximum Marks: 24

Note: Note: Answer any EIGHT questions from this section. Each question carries 3 marks.

Q2. Define: (iii) (ii) (i) :

(i) Tyndall effect

(ii) Valency

(iii) Electronegativity

Q3. Write down three differences between Alkali metals and Alkaline earth metals.

Q4. What is an aqueous solution? Give an example. Air is a mixture of gases then why Nitrogen gas is considered solvent?

Q5. What are isotopes? Describe isotopes of Chlorine. (Isotopes)

Q6. Identify group and period in the period table of the following elements: (i) 3Li(ii) :

(i) 3Li

(ii) 6C

(iii) 13Al 6C (iii) 13Al

Q7. Write down any three points of atomic model proposed by Schrodinger. 100cm3 (C2H2O4) 2M

Q8. What is the mass of Oxalic acid (C2H2O4) required to prepare 100cm3 volume of 2M Oxalic acid solution? (where Atomic masses: C = 12, H (C=12 H=1 O=16 :) = 1, O = 16)

Q9. Balance the following chemical equations:

(iii) $NH_3 + O_2 \rightarrow NO +$

(i) $Ca + O_2 \rightarrow CaO$

(ii) $KClO_3 \rightarrow KCl + O_2$ H_2O iii) $NH_3 + O_2 \rightarrow NO + H_2O$) ii) $KClO_3 \rightarrow KCl + O_2$) i) $Ca + O_2 \rightarrow CaO$

Q10. State Faraday's 1 law of electrolysis and explain it

Q11. 60ml sample of a gas is heated form 27°C to 52°C, at constant C°52 C°27 60ml pressure. What will be final volume of this gas?

Q12. Write down the chemical formulae of given compounds: (ii) (i) :

(i) Lime stone

(ii) Washing soda

(iii) Sugar (iii)

Q13. Give reason: (ii) (i) :

(i) Why gasoline does not dissolve

(ii) Alkali metals are not found in

SECTION “C” : (Detailed –Answer Question)

Maximum Marks: 12

Note: Note: Attempt any TWO (2) questions from this section. Each question carries 6 marks.

Q14. Define Chemistry, and list any five branches of it and explain them.

Q15. Define ionic bond. Illustrate the bond formation in Sodium Chloride (NaCl) (NaCl) also write any two characteristics of ionic compounds. OR Describe discharge tube experiment with labelled diagram and state what were the justification and conclusion of J.J. Thomson in this respect

Q16. What is electroplating? How can a steel spoon be electroplated with chromium explain with electrodes reaction also draw a labelled diagram of electrolytic cell.